

#### 4.5.2 The photoelectric effect

| Learning outcomes                                                                             |                                                                                                                                                     | Additional guidance                                                                                                                                                                                           |
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| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i> |                                                                                                                                                     |                                                                                                                                                                                                               |
| (a)                                                                                           | (i) photoelectric effect, including a simple experiment to demonstrate this effect                                                                  | Learners should understand that the photoelectric effect provides evidence for particulate nature of electromagnetic radiation.<br>HSW1, 2, 3, 7, 11 Internet research on the development of quantum physics. |
|                                                                                               | (ii) demonstration of the photoelectric effect using, e.g. gold-leaf electroscope and zinc plate                                                    |                                                                                                                                                                                                               |
| (b)                                                                                           | a one-to-one interaction between a photon and a surface electron                                                                                    |                                                                                                                                                                                                               |
| (c)                                                                                           | Einstein's photoelectric equation $hf = \phi + KE_{\max}$                                                                                           | M2.3                                                                                                                                                                                                          |
| (d)                                                                                           | work function; threshold frequency                                                                                                                  |                                                                                                                                                                                                               |
| (e)                                                                                           | the idea that the maximum kinetic energy of the photoelectrons is independent of the intensity of the incident radiation                            |                                                                                                                                                                                                               |
| (f)                                                                                           | the idea that rate of emission of photoelectrons above the threshold frequency is directly proportional to the intensity of the incident radiation. |                                                                                                                                                                                                               |